

ADITYA RAJ

B.Tech CSE Student | Amrita Vishwa Vidyapeetham, Haridwar

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EDUCATION

Amrita Vishwa Vidyapeetham, Haridwar

2025 – 2029

B.Tech – Computer Science & Engineering | CGPA: 8.0 (Semester 1)

Anjabi Singh College, Bihar Board

2022 – 2024

Class XII (Intermediate) | Percentage: 72%

The Divine Public School

Completed 2022

2Class X | Percentage: 86%

TECHNICAL SKILLS

Languages

C, C++, Python, Java, JavaScript

Web Technologies

HTML, CSS

CS Fundamentals

Data Structures & Algorithms (DSA)

Tools & Platforms

Arduino, MATLAB, Simulink, Proteus, Tinkercad, NCviewer, NodeMCU (ESP8266)

Concepts

OOP, Graph Theory, Combinatorics, Discrete Mathematics, Mesh Analysis

PROJECTS

Smart Traffic Control System with Street Light

2025–26

Arduino UNO R4 WiFi | IoT | C++ / HTML

- Designed a 4-road smart intersection controller using a Master-Slave Arduino architecture over WiFi (802.11 b/g/n).
- Integrated IR, Ultrasonic, LDR, and RFID (MFRC522) sensors with SSD1306 OLED display and a REST HTTP API for real-time web control.

Autonomous Vehicle Navigation System Optimization Using Graph Theory

2025–26

Discrete Mathematics / Graph Theory | C++

- Modelled road networks as weighted directed graphs ($G = (V, E)$) incorporating distance, traffic density, accident risk, and energy consumption.
- Applied Dijkstra, A*, and Kruskal algorithms to optimise travel time, congestion avoidance, and multi-vehicle coordination simultaneously.

Automated Timetable Generator Using Combinatorial & Recursive Techniques

2025–26

Discrete Mathematics | Python

- Formulated timetable scheduling as a combinatorial optimisation problem and solved it via recursive backtracking.
- Eliminated teacher/room conflicts and met subject-frequency requirements for an entire academic semester automatically.

Electrical Circuit Mesh Analysis Using Inverse Matrix ($Ax = B$)

2025–26

Linear Algebra | MATLAB

- Applied Kirchhoff's Voltage Law (KVL) to multi-mesh circuits and converted simultaneous equations into matrix form.
- Solved for unknown mesh currents using MATLAB's matrix-inverse method, validating results analytically.

Library Management System Using OOP (Java)

2025–26

OOP | Java

- Implemented a Java-based LMS supporting book addition, member registration, and issue/return workflows using OOP principles (encapsulation, abstraction, association).

Smart Light Detection & Alert System (LDR + NodeMCU)

2025–26

OOP | Java | IoT | ESP8266

- Built an LDR-based light intensity detector with LED and buzzer control on NodeMCU ESP8266, modelled in Java OOP for system logic.

Closed-Loop RPM Control System for DC Motor

2025–26

Manufacturing Project | Arduino | Proteus ISIS

- Designed a PID-controlled DC motor speed regulation system using Arduino UNO, 4N25 optocoupler, and L298 motor driver, simulated in Proteus ISIS with 16×2 LCD display.

MATLAB Simulation of Blackbody Radiation & Wien's Displacement Law

2025–26

Physics | MATLAB

- Simulated Planck's radiation law across multiple temperatures, plotted Intensity vs. Wavelength graphs, and verified Wien's Displacement Law numerically.

CNC Rectangular Block Cutting – G-Code & M-Code Programming

2025–26

Manufacturing | CNC | NCviewer

- Programmed a complete CNC milling routine (ISO 6983 standard) for a 100×60×10 mm aluminium block with G-Code/M-Code, verified with NCviewer simulation.

Pneumatic OR Logic Circuit (Shuttle Valve) – Simulink Simscape

2025–26

MATLAB/Simulink | Simscape Fluids

- Designed and simulated a dual push-button pneumatic OR logic circuit using a shuttle valve in MATLAB/Simulink Simscape Fluids, validating pressure propagation and actuator response.

INTERESTS & HOBBIES

Sports: Cricket, Badminton, Volleyball **Other:** Watching Movies, Travelling